

# UE Apprentissage SIA S9

Supervise Learning

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# Outline

Introduction

Introductory Example

Multi layer Perceptron

Convolutional Neural Networks

Transformer

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- CV:

2004-2007 Ph.D. degree in Signal and Image Processing from the INP, Grenoble & the University of Iceland

2007-2008 Assistant Professor Grenoble

2008-2010 Post-doc position at INRIA - MISTIS Team

2010-2011 Assistant Professor Toulouse

2011-2018 Associate Professor at DYNAFOR & INP, Toulouse

Since 2018 Research (CR1) at CESBIO, INRAe

- Research interests are:

Machine learning for environmental/ecological monitoring

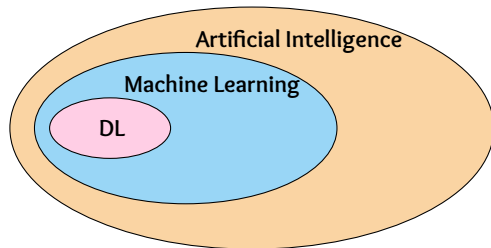
# Course Objectives

TBC

# What is Supervised (Machine) Learning ?

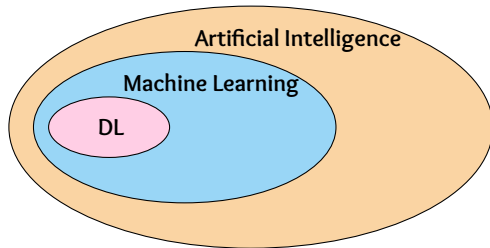
- **Artificial Intelligence**

*Perform human tasks using computers and algorithms*



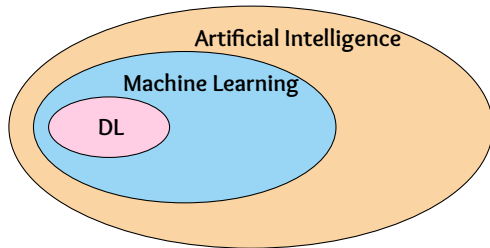
# What is Supervised (Machine) Learning ?

- **Artificial Intelligence**  
*Perform human tasks using computers and algorithms*
- Mitchell, *The discipline of machine learning*:  
*Machine Learning is defined as the capacity of a computer program to improve its performance measure with observations*



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- Mitchell, *The discipline of machine learning*:  
*Machine Learning is defined as the capacity of a computer program to improve its performance measure with observations*
- **Deep Learning**
  - ★ Lot of data data
  - ★ Complex model
  - ★ High performance computing





# Main notations

## Data

- Observed data  $\mathbf{x} \in \mathbb{R}^d$ , called *input variables* or *predictors*.
- Data to be predicted  $y \in \mathbb{R}^p$ , called *output variables* or *responses*.

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## Learning Problem

- If  $y$  are quantitative data: *regression* and  $y \in \mathbb{R}^p$ .
- If  $y$  are categorical data: *classification* and  $y \in \{C_1, \dots, C_C\}$  with  $C_i$  refers to the  $i^{th}$  category.

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## Prediction function

$$\begin{aligned} f_{\theta} : \mathbb{R}^d &\rightarrow \mathbb{R}^p \\ \mathbf{x} &\mapsto y \end{aligned}$$

# Online References

- Aston Zhang et al. “Dive into Deep Learning”. In: *arXiv preprint arXiv:2106.11342* (2021)
- Simon J.D. Prince. *Understanding Deep Learning*. MIT Press, 2023. URL: <https://udlbook.github.io/udlbook/>
- Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: <http://probml.github.io/book1>
- Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2>

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# 1D linear model

- $f(x) = wx + b$  and  $\theta = (w, b)$
- Loss function:  $\ell(f(x_i), y_i) = (f(x_i) - y_i)^2$
- Objective function:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i$$

- Gradients update:  $\theta^{t+1} = \theta^t - \eta \nabla_{\theta} \mathcal{L}$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{2}{n} \sum_{i=1}^n x_i (wx_i + b - y_i)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{2}{n} \sum_{i=1}^n (wx_i + b - y_i)$$

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## Work 2.2:

Implement the 1D regression in pytorch of the following function:

$$f(x) = \frac{1}{2}x + \frac{1}{2} \sin\left(\frac{2\pi}{5}x\right)$$

Generate 100 pairs of random samples  $(x_i, \tilde{f}_{x_i})$  between  $[-5, 5]$  such as  $\tilde{f}(x_i) = f(x_i) + \epsilon_i$ ,  $\epsilon_i \sim \mathcal{N}(0, 0.1)$

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Introductory Example

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Convolutional Neural Networks

Transformer

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Introduction

Introductory Example

Multi layer Perceptron

Convolutional Neural Networks

Transformer

# Bibliography I

- [1] T. Mitchell. *The discipline of machine learning*. Tech. rep. CMU ML-06 108. 2006. URL: <http://ra.adm.cs.cmu.edu/anon/usr0/ftp/anon/ml/CMU-ML-06-108.pdf>.
- [2] Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2>.
- [3] Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: <http://probml.github.io/book1>.
- [4] Simon J.D. Prince. *Understanding Deep Learning*. MIT Press, 2023. URL: <https://udlbook.github.io/udlbook/>.
- [5] Aston Zhang et al. “Dive into Deep Learning”. In: *arXiv preprint arXiv:2106.11342* (2021).

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